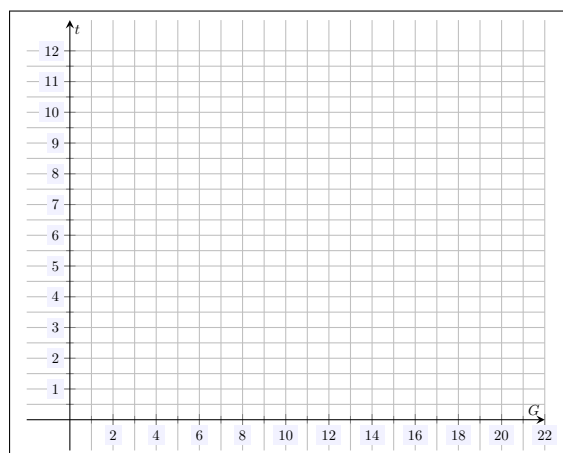


Recitation: Linear Functions Review

Problem 1. A small tanker truck is depositing its gas at a storage facility. The tanker is carrying 11,600 gallons of gas and is emptying its tank at a rate of 528.3 gal/min. Let $G(t)$ denote the volume of gas, in thousands, left in the tanker t minutes from now.

- (a) Explain why $G(t)$ is linear.
- (b) Find $G(t)$ and sketch it in the plot below.
- (c) Interpret the slope of $G(t)$.
- (d) Interpret the y -intercept for $G(t)$.
- (e) Find and interpret (if possible) the x -intercept for $G(t)$.



Problem 2. Find the equation of the line with x -intercept 5 and y -intercept -6 .

Problem 3. Consider the linear function $\ell(x) = \frac{3x-5}{4}$.

- (a) Find the slope and y -intercept for $\ell(x)$.
- (b) Find the x -intercept(s) of $\ell(x)$.
- (c) Is $(-1, 2)$ on the graph of $\ell(x)$?
- (d) Compute $\ell(1)$.
- (e) Determine the domain and range for $\ell(x)$.